

Generalized Harmonic Progression Part II

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Abstract

In a previous paper, we saw how to create formulae for the sum of the terms of a harmonic progression of order k , $HP_k(n)$, with integer parameters, a and b . In this new paper we make those formulae even more general by lifting the restriction that the parameters be integers. The new formula holds always, except when $ib/a \in \mathbb{Z}$. Here a slightly modified version of the reasoning used before is introduced, in a standalone exposition that does not require prior knowledge of the precursor paper.

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1 Introduction

This manuscript demonstrates how to produce formula for a harmonic progression of order k with complex parameters,

$$\text{HP}_k(n) = \sum_{j=1}^n \frac{1}{(a\mathbf{i}j + b)^k},$$

with a an integer, b a complex number and $\mathbf{i}$ the imaginary unit.

The motivation to create such a formula came from the realization that, unlike the sum of $1/(j^2 - 1)$ over j , a sum such as $1/(j^2 + 1)$ can not be derived from the formulae created in [4].

If a formula for $\text{HP}_k(n)$ with integer a and complex b can be produced, formulae for most, if not all, sums of the type $\sum_j 1/p(j)$, with $p(j)$ a polynomial in j with complex coefficients, can be produced. Besides, $\text{HP}_k(n)$ provides the partial sums of the Hurwitz zeta function, $\zeta(k, b)$, at the positive integers k , whose formula can be obtained by taking the limit of $\text{HP}_k(n)$ when n goes to infinity. With a little more effort, it may even be possible to obtain the analytic continuation of $\zeta(k, b)$.

The choice of integer a and complex b is good enough in most of the cases (since one can always make a integer by dividing the parameters by a), and when it is not, one can resort to the formulae from the previous paper [4], which assume a and b to be integers.

The method makes use again of the Faulhaber formula [2] for the sum of the powers of the first n positive integers, which are given by,

$$\sum_{j=1}^n j^{2i} = \frac{n^{2i+1}}{2i+1} + \sum_{j=0}^i \frac{(2i+1)! B_{2j}}{(2j)!(2i+1-2j)!} n^{2i+1-2j} \quad (1)$$

$$\sum_{j=1}^n j^{2i+1} = \frac{n^{2i+2}}{2i+2} + \sum_{j=0}^i \frac{(2i+2)! B_{2j}}{(2j)!(2i+2-2j)!} n^{2i+2-2j}, \quad (2)$$

where B_{2j} are the non-zero Bernoulli numbers.

2 Review of preview results

Before we begin, let us recapitulate some results from [4] for a bit of context. We had seen that the sum of the terms of a generalized harmonic progression with integer

parameters, a and b , for even and odd powers is given, respectively, by,

$$\sum_{j=1}^n \frac{1}{(aj+b)^{2k}} = -\frac{1}{2b^{2k}} + \frac{1}{2(an+b)^{2k}} - \frac{(-1)^k (2\pi)^{2k}}{2} \int_0^1 \sum_{j=0}^k \frac{B_{2j} (2-2^{2j}) (1-u)^{2k-2j}}{(2j)!(2k-2j)!} (\sin 2\pi(an+b)u - \sin 2\pi bu) \cot \pi au \, du,$$

and

$$\sum_{j=1}^n \frac{1}{(aj+b)^{2k+1}} = -\frac{1}{2b^{2k+1}} + \frac{1}{2(an+b)^{2k+1}} - \frac{(-1)^k (2\pi)^{2k+1}}{2} \int_0^1 \sum_{j=0}^k \frac{B_{2j} (2-2^{2j}) (1-u)^{2k+1-2j}}{(2j)!(2k+1-2j)!} (\cos 2\pi(an+b)u - \cos 2\pi bu) \cot \pi au \, du$$

Notice that there may be singularities on both sides of these equations if, for example, $aj+b=0$ for some $j \leq n$ (that is, a divides b and they have different signs). In those cases, if the singularities are ignored wherever they occur the formula still holds.

This means that, for instance, if $b=0$ then the above formulae reduce to the formulae of the generalized harmonic numbers from reference [3].

To see why singularities can be removed, let us look at the formula for the odd powers. From that formula, it is easy to see why it still holds when singularities are ignored. For example, let us take the forward difference of $\text{HP}(n)$,

$$2\pi \int_0^1 (1-u) (\cos 2\pi(an+b)u - \cos 2\pi(a(n-1)+b)u) \cot \pi a(1-u) \, du = -\frac{1}{an+b} - \frac{1}{a(n-1)+b}$$

Now, if for instance $an+b=0$, the left-hand side of the above equation yields $1/a$, even though $1/(an+b)$ is a singularity on the right-hand side. Therefore, by ignoring the singularity, the equation still holds.

There was also a unified formula based on the exponential (that is, the initial equation).

For every integer a , b and integer $k \geq 1$:

$$\sum_{j=1}^n \frac{1}{(aj+b)^k} = -\frac{1}{2b^k} + \frac{1}{2(an+b)^k} + \frac{i(2\pi i)^k}{2} \int_0^1 \sum_{j=0}^k \frac{B_j(1-u)^{k-j}}{j!(k-j)!} \left(e^{2\pi i(an+b)u} - e^{2\pi i b u} \right) \cot \pi a u \, du \quad (3)$$

3 Approach based on exponential

The reasoning to build a formula for the harmonic progression relies on the Taylor series expansion of the exponential, and seizes upon the fact that it is constant when a is integer and b is complex:

$$e^{2\pi i(aj+b)} = e^{2\pi b} \Rightarrow \sum_{i=0}^{\infty} \frac{(2\pi i(aj+b))^i}{i!} = e^{2\pi b}, \text{ if } a \text{ is integer} \quad (4)$$

Starting from the above initial equation, one ends up with the below after a few steps:

$$\frac{1}{a i j + b} (e^{2\pi b} - 1) = 2\pi \sum_{i=0}^{\infty} \left(\frac{(2\pi i(aj+b))^{2i}}{(2i+1)!} + \frac{(2\pi i(aj+b))^{2i+1}}{(2i+2)!} \right) \quad (5)$$

Note there are now two power series, as opposed to one previously (that is, in [3] and [4]).

At this point it is evident that this formula does not work if $e^{2\pi b} = 1$, which happens when b is a purely complex number with integer imaginary part ($\Re(b) = 0$ and $\Im(b) \in \mathbb{Z}$). In most situations, this issue can be avoided by making b non-integer, such as by dividing b by a . Likewise, the formula does not apply if a is not integer ($a \notin \mathbb{Z}$). In that case a can be made integer by dividing both parameters by a :

$$\sum_{j=1}^n \frac{1}{a i j + b} = \frac{1}{a} \sum_{j=1}^n \frac{1}{i j + b/a}$$

However, in both cases this solution does not work if $i b/a$ is integer, including zero. In those cases the previous formulae are applicable.

3.1 Lagrange's trigonometric identities

Through the Lagrange's identities, which stem from the sum of the terms of a geometric progression, functions can be created whose power series take similar forms to the power series that are obtained when the initial equation from the previous section is expanded.

For the complex harmonic progression, these functions are the same as the ones used in [4], with only a slight transformation ($a \mapsto a\mathbf{i}$). Nonetheless, they are obtained by means of the Lagrange's trigonometric identities,

$$\sum_{j=1}^k \cos \frac{2\pi n(a\mathbf{i}j + b)}{k} = -\frac{1}{2} \cos \frac{2\pi bn}{k} + \frac{1}{2} \cos 2\pi n \left(a\mathbf{i} + \frac{b}{k} \right) + \cos \pi n \left(a\mathbf{i} + \frac{2b}{k} \right) \sin \pi a\mathbf{i}n \cot \frac{\pi a\mathbf{i}n}{k}, \quad (6)$$

and

$$\sum_{j=1}^k \sin \frac{2\pi n(a\mathbf{i}j + b)}{k} = -\frac{1}{2} \sin \frac{2\pi bn}{k} + \frac{1}{2} \sin 2\pi n \left(a\mathbf{i} + \frac{b}{k} \right) + \sin \pi n \left(a\mathbf{i} + \frac{2b}{k} \right) \sin \pi a\mathbf{i}n \cot \frac{\pi a\mathbf{i}n}{k}, \quad (7)$$

along with the cosine and sine addition formulae [1]:

$$\cos(x + y) = \cos x \cos y - \sin x \sin y, \text{ and } \sin(x + y) = \sin x \cos y + \cos x \sin y$$

Power series for the left-hand side of the Lagrange identities can be derived with the employment of (1) and (2), resulting in the following power series for each function on the right-hand side:

$$\cos 2\pi n \left(a\mathbf{i} + \frac{b}{k} \right) = \sum_{i=0}^{\infty} (-1)^i \left(\frac{2\pi bn}{k} \right)^{2i} \left(\sum_{j=0}^i \frac{(a\mathbf{i}k/b)^{2j}}{(2j)!(2i-2j)!} + \sum_{j=0}^{i-1} \frac{(a\mathbf{i}k/b)^{2j+1}}{(2j+1)!(2i-2j-1)!} \right)$$

$$\sin 2\pi n \left(a\mathbf{i} + \frac{b}{k} \right) = \sum_{i=0}^{\infty} (-1)^i \left(\frac{2\pi bn}{k} \right)^{2i+1} \left(\sum_{j=0}^i \frac{(a\mathbf{i}k/b)^{2j}}{(2j)!(2i+1-2j)!} + \sum_{j=0}^i \frac{(a\mathbf{i}k/b)^{2j+1}}{(2j+1)!(2i-2j)!} \right)$$

$$\begin{aligned} & \cos \pi n \left(a\mathbf{i} + \frac{2b}{k} \right) \sin \pi a\mathbf{i}n \cot \frac{\pi a\mathbf{i}n}{k} = \\ & \sum_{i=0}^{\infty} (-1)^i \left(\frac{2\pi bn}{k} \right)^{2i} \left(\sum_{j=0}^i \frac{(a\mathbf{i}/b)^{2j}}{(2i-2j)!} \sum_{q=0}^j \frac{B_{2q} k^{2j+1-2q}}{(2j+1-2q)!(2q)!} + \sum_{j=0}^{i-1} \frac{(a\mathbf{i}/b)^{2j+1}}{(2i-2j-1)!} \sum_{q=0}^j \frac{B_{2q} k^{2j+2-2q}}{(2j+2-2q)!(2q)!} \right) \\ & \sin \pi n \left(a\mathbf{i} + \frac{2b}{k} \right) \sin \pi a\mathbf{i}n \cot \frac{\pi a\mathbf{i}n}{k} = \\ & \sum_{i=0}^{\infty} (-1)^i \left(\frac{2\pi bn}{k} \right)^{2i+1} \left(\sum_{j=0}^i \frac{(a\mathbf{i}/b)^{2j}}{(2i+1-2j)!} \sum_{q=0}^j \frac{B_{2q} k^{2j+1-2q}}{(2j+1-2q)!(2q)!} + \sum_{j=0}^i \frac{(a\mathbf{i}/b)^{2j+1}}{(2i-2j)!} \sum_{q=0}^j \frac{B_{2q} k^{2j+2-2q}}{(2j+2-2q)!(2q)!} \right) \end{aligned}$$

3.2 Harmonic progression

To obtain $HP(n)$, equation (5) is summed over j , and then $(aij + b)^{2i}$ and $(aij + b)^{2i+1}$ are expanded using the binomial theorem, and lastly the sums of j^{2q} and j^{2q+1} over j are replaced with their respective Faulhaber formulae. After all the possible simplifications are carried out, one ends up with a few power series, shown below.

First, one has the independent term given by:

$$-\pi \sum_{i=0}^{\infty} \left(\frac{(2\pi b)^{2i}}{(2i+1)!} + \frac{(2\pi b)^{2i+1}}{(2i+2)!} \right) = -\frac{e^{2\pi b} - 1}{2b}$$

For the other, more complicated power series, integral representations can be obtained with the aid of their corresponding functions from section (3.1):

$$\begin{aligned} \pi \sum_{i=0}^{\infty} \frac{(2\pi b)^{2i}}{2i+1} \left(\sum_{j=0}^i \frac{(ain/b)^{2j}}{(2j)!(2i-2j)!} + \sum_{j=0}^{i-1} \frac{(ain/b)^{2j+1}}{(2j+1)!(2i-2j-1)!} \right) &= -\frac{i\pi}{n} \int_0^{in} \cos 2\pi x \left(ai + \frac{b}{n} \right) dx \\ \pi \sum_{i=0}^{\infty} \frac{(2\pi b)^{2i+1}}{2i+2} \left(\sum_{j=0}^i \frac{(ain/b)^{2j}}{(2j)!(2i+1-2j)!} + \sum_{j=0}^i \frac{(ain/b)^{2j+1}}{(2j+1)!(2i-2j)!} \right) &= -\frac{\pi}{n} \int_0^{in} \sin 2\pi x \left(ai + \frac{b}{n} \right) dx \\ 2\pi \sum_{i=0}^{\infty} \frac{(2\pi b)^{2i}}{2i+1} \left(\sum_{j=0}^i \frac{(ai/b)^{2j}}{(2i-2j)!} \sum_{q=0}^j \frac{B_{2q} n^{2j+1-2q}}{(2j+1-2q)!(2q)!} + \sum_{j=0}^{i-1} \frac{(ai/b)^{2j+1}}{(2i-2j-1)!} \sum_{q=0}^j \frac{B_{2q} n^{2j+2-2q}}{(2j+2-2q)!(2q)!} \right) \\ &= -\frac{2\pi i}{n} \int_0^{in} \cos \pi x \left(ai + \frac{2b}{n} \right) \sin(\pi a i x) \cot \frac{\pi a i x}{n} dx \\ 2\pi \sum_{i=0}^{\infty} \frac{(2\pi b)^{2i+1}}{2i+2} \left(\sum_{j=0}^i \frac{(ai/b)^{2j}}{(2i+1-2j)!} \sum_{q=0}^j \frac{B_{2q} n^{2j+1-2q}}{(2j+1-2q)!(2q)!} + \sum_{j=0}^i \frac{(ai/b)^{2j+1}}{(2i-2j)!} \sum_{q=0}^j \frac{B_{2q} n^{2j+2-2q}}{(2j+2-2q)!(2q)!} \right) \\ &= -\frac{2\pi}{n} \int_0^{in} \sin \pi x \left(ai + \frac{2b}{n} \right) \sin(\pi a i x) \cot \frac{\pi a i x}{n} dx \end{aligned}$$

Therefore, after putting it all together and making a change of variables, the formula below is found, which holds as long as a is integer and ib is not integer:

$$\sum_{j=1}^n \frac{1}{aij + b} = -\frac{1}{2b} + \frac{1}{2(ain + b)} + \frac{2\pi}{e^{2\pi b} - 1} \int_0^1 e^{\pi(ain+2b)u} \sin \pi a n u \cot \pi a u du \quad (8)$$

In this formula, the independent term came from the first power series, the second term from the next two power series and the third term from the last two power series, which is somewhat evident.

As discussed before, provided that $ib/a \notin \mathbb{Z}$, the restriction that a be integer can be lifted by transforming the formula, for example, as follows,

$$\frac{1}{a} \sum_{j=1}^n \frac{1}{ij + b/a} = -\frac{1}{2b} + \frac{1}{2(ain + b)} + \frac{2\pi}{a(e^{2\pi b/a} - 1)} \int_0^1 e^{\pi(in+2b/a)u} \sin \pi nu \cot \pi u \, du, \quad (9)$$

which holds for any $a, b \in \mathbb{C}$. There are endless ways of doing this, this example is just one of the possibilities.

3.3 Generalization

To generalize the previous formula to $\text{HP}_k(n)$, first note that a recurrence is obtained at each new step. For example, for the third power one has:

$$(e^{2\pi b} - 1) \sum_{j=1}^n \frac{1}{(aij + b)^3} = \sum_{j=1}^n \left(\frac{(2\pi)^2}{2!} \frac{1}{aij + b} + \frac{2\pi}{1!} \frac{1}{(aij + b)^2} + (2\pi)^3 \sum_{i=0}^{\infty} \frac{(2\pi(aij + b))^{2i}}{(2i + 3)!} + \frac{(2\pi(aij + b))^{2i+1}}{(2i + 4)!} \right)$$

This means that only the last sum in the equation above is complicated. As before, this recurrence is such that in the final formula the terms that go outside of the integral reduce to very simple forms, with the integral being the only challenging part.

Let $p_k(u)$ be the polynomial in u that goes within the integral. Then for integer a and non-integer ib/a ,

$$\sum_{j=1}^n \frac{1}{(aij + b)^k} = -\frac{1}{2b^k} + \frac{1}{2(ain + b)^k} + (2\pi)^k \int_0^1 p_k(u) e^{\pi(ain+2b)u} \sin \pi au \cot \pi au \, du, \quad (10)$$

where $p_k(u)$ is given by the recurrence equation,

$$(e^{2\pi b} - 1)p_k(u) = \begin{cases} 1, & \text{if } k = 1 \\ \frac{(1-u)^{k-1}}{(k-1)!} + \sum_{j=1}^{k-1} \frac{p_j(u)}{(k-j)!}, & \text{if integer } k > 1 \end{cases}$$

and where term $(1-u)^{k-1}/(k-1)!$ came from the observation of the patterns.

Now to solve the recurrence, let $p(x)$ be the generating function of $p_k(u)$. By analyzing the recurrence, it is possible to conclude that:

$$(e^{2\pi b} - 1)p(x) - (-1 + e^x)p(x) = x e^{(1-u)x} \Rightarrow p(x) = -\frac{x e^{(1-u)x}}{e^x - e^{2\pi b}} \quad (11)$$

The solution of this equation is not trivial, as the general term of the power series of $p(x)$ is not simple.

To find $p_k(u)$, let δ_{ij} be the Kronecker delta ($\delta_{ij} = 1$, if $i = j$, $\delta_{ij} = 0$ otherwise), and $\text{Li}_k(z)$, the polylogarithm^[1].

The $p_k(u)$ are the coefficients of the Taylor series expansion of $p(x)$, which are,

$$p_k(u) = \frac{p^{(k)}(0)}{k!} = \sum_{j=1}^k \frac{(\delta_{1j} + \text{Li}_{-j+1}(e^{-2\pi b})) (1-u)^{k-j}}{(j-1)!(k-j)!},$$

and the identity below,

$$\sum_{j=1}^k \frac{\text{Li}_{-j+1}(e^{-2\pi b})(1-u)^{k-j}}{(j-1)!(k-j)!} = \frac{e^{-2\pi b}}{\Gamma(k)} \Phi(e^{-2\pi b}, -k+1, 2-u),$$

provides an extension of the summation to non-integer k .

Further simplifications can be attained through the identity below,

$$(1-u)^{k-1} + e^{-2\pi b} \Phi(e^{-2\pi b}, -k+1, 2-u) = \Phi(e^{-2\pi b}, -k+1, 1-u), \quad (12)$$

that stems from the definition of the Lerch Φ function and also through Euler's formula:

$$e^{\pi i a n u} \sin \pi a n u = (\cos \pi a n u + i \sin \pi a n u) \sin \pi a n u = \frac{i}{2} (1 - e^{2\pi i a n u}), \quad (13)$$

Therefore, if k is complex with $\Re(k) > 0$ and a is integer, one has,

$$\begin{aligned} \sum_{j=1}^n \frac{1}{(a i j + b)^k} &= -\frac{1}{2b^k} + \frac{1}{2(a i n + b)^k} \\ &\quad - \frac{i(2\pi)^k}{2\Gamma(k)} \int_0^1 e^{-2\pi b(1-u)} \Phi(e^{-2\pi b}, -k+1, 1-u) (e^{2\pi i a n u} - 1) \cot \pi a u \, du \end{aligned} \quad (14)$$

Finally, the formula can be turned into a more straightforward form by making the offset parameter equal to $\mathbf{i}b$:

$$\sum_{j=1}^n \frac{1}{(aj+b)^k} = -\frac{1}{2b^k} + \frac{1}{2(an+b)^k} - \frac{\mathbf{i}(2\pi\mathbf{i})^k}{2\Gamma(k)} \int_0^1 e^{-2\pi\mathbf{i}b(1-u)} \Phi(e^{-2\pi\mathbf{i}b}, -k+1, 1-u) (e^{2\pi\mathbf{i}anu} - 1) \cot \pi au \, du \quad (15)$$

3.4 Solution of the transform

A transformation of (10) is shown below, which is just setting $a = 1$ and $b \mapsto b/a$ and dividing both sides by a^k , and which can be more useful for the purpose of finding the limit of the harmonic progression of order k :

$$\sum_{j=1}^n \frac{1}{(a\mathbf{i}j+b)^k} = -\frac{1}{2b^k} + \frac{1}{2(a\mathbf{i}n+b)^k} + \left(\frac{2\pi}{a}\right)^k \int_0^1 p_k(u) e^{\pi(\mathbf{i}n+2b/a)u} \sin \pi nu \cot \pi u \, du \quad (16)$$

Now, the generating function of the $p_k(u)$ becomes:

$$p(x) = -\frac{x e^{x(1-u)/a}}{a(e^{x/a} - e^{2\pi b/a})} \Rightarrow \frac{p^{(k)}(0)}{k!} = \frac{e^{-2\pi b/a}}{a^k(k-1)!} \Phi(e^{-2\pi b/a}, -k+1, 1-u),$$

Therefore, if $\Re(k) > 0$ and $\mathbf{i}b/a \notin \mathbb{Z}$, one has:

$$\sum_{j=1}^n \frac{1}{(a\mathbf{i}j+b)^k} = -\frac{1}{2b^k} + \frac{1}{2(a\mathbf{i}n+b)^k} - \frac{\mathbf{i}}{2\Gamma(k)} \left(\frac{2\pi}{a}\right)^k \int_0^1 e^{-2\pi(1-u)b/a} \Phi(e^{-2\pi b/a}, -k+1, 1-u) (e^{2\pi\mathbf{i}nu} - 1) \cot \pi u \, du \quad (17)$$

The formula can be simplified further by letting $a = -\mathbf{i}$. Thus, if $\Re(k) > 0$ and $b \notin \mathbb{Z}$ then:

$$\sum_{j=1}^n \frac{1}{(j+b)^k} = -\frac{1}{2b^k} + \frac{1}{2(n+b)^k} - \frac{\mathbf{i}(2\pi\mathbf{i})^k}{2\Gamma(k)} \int_0^1 e^{-2\pi\mathbf{i}b(1-u)} \Phi(e^{-2\pi\mathbf{i}b}, -k+1, 1-u) (e^{2\pi\mathbf{i}nu} - 1) \cot \pi u \, du \quad (18)$$

3.5 Application examples

Obtaining formulae for the sum of the reciprocal of a polynomial $p(j)$ of order k over j is straightforward as long as the decomposition of the polynomial does not have any multiple roots. If x_q , $q = 1, \dots, k$, are the k unique roots of $p(j)$ then:

$$\frac{1}{p(j)} = \sum_{q=1}^k \frac{c_q}{j - x_q} \quad (19)$$

If any of the roots is multiple, then the decomposition is a little more complicated, but the literature has a general method to find the c_q as a function of $p(j)$, the roots x_q and their multiplicity, anyway.

As a particular case, to obtain a formula for the sum of $1/(j^{2k} + 1)$ over j it is necessary to find the $2k$ complex roots of $x^{2k} + 1 = 0$, as below:

$$\sum_{q=1}^{2k} \frac{c_q}{j - x_q} = \frac{1}{j^{2k} + 1} \quad (20)$$

Let us see two examples on how the newly found formulae can be applied to obtain sums that were not possible with the previous formulae created in [4].

The first example is the sum of $1/(j^2 + 1)$ over j . The first step to derive this formula is to decompose it into a linear combination as below:

$$\frac{1}{j^2 + 1} = \frac{1}{2(\mathbf{i}j + 1)} - \frac{1}{2(\mathbf{i}j - 1)}$$

After a few simplifications, the formula below is found:

$$\sum_{j=1}^n \frac{1}{j^2 + 1} = -\frac{1}{2} + \frac{1}{2(n^2 + 1)} + \pi \operatorname{csch} 2\pi \int_0^1 \cosh 2\pi(2u - 1) \sin 4\pi n u \cot 2\pi u \, du \quad (21)$$

The second example is another quadratic polynomial, which can be decomposed as below,

$$\frac{1}{j^2 + 2j + 2} = \frac{1}{2(\mathbf{i}j + 1 + \mathbf{i})} - \frac{1}{2(\mathbf{i}j - 1 + \mathbf{i})},$$

and which therefore implies the formula:

$$\begin{aligned} \sum_{j=1}^n \frac{1}{j^2 + 2j + 2} &= -\frac{1}{4} + \frac{1}{2(n^2 + 2n + 2)} \\ &+ \frac{2\pi}{e^{4\pi} - 1} \int_0^1 (e^{4\pi(1-u)} + e^{4\pi u}) \cos 2\pi(n+2)u \sin 2\pi n(1-u) \cot 2\pi(1-u) \, du \end{aligned} \quad (22)$$

Therefore, it becomes evident that the real problem is finding the linear combination that yields the reciprocal of the polynomial that one is interested in.

4 Approaches based on cosine and sine

Though the exponential-based approach is probably more useful, if $\cos 2\pi(j+b) = \cos 2\pi b$ and $\sin 2\pi(j+b) = \sin 2\pi b$ are used as initial identities, different formulae can be obtained.

For the cosine, starting from,

$$\cos 2\pi(j+b) = \cos 2\pi b = \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i}}{(2i)!},$$

produces the recurrence equations:

$$\begin{cases} \frac{2(\sin \pi b)^2}{(j+b)} = 2\pi \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i+1}}{(2i+2)!} \\ \frac{2(\sin \pi b)^2}{(j+b)^2} = (2\pi)^2 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i}}{(2i+2)!} \\ \frac{2(\sin \pi b)^2}{(j+b)^3} = \frac{(2\pi)^2}{2!(j+b)} - (2\pi)^3 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i+1}}{(2i+4)!} \\ \frac{2(\sin \pi b)^2}{(j+b)^4} = \frac{(2\pi)^2}{2!(j+b)^2} - (2\pi)^4 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i}}{(2i+4)!} \\ \frac{2(\sin \pi b)^2}{(j+b)^5} = \frac{(2\pi)^2}{2!(j+b)^3} - \frac{(2\pi)^4}{4!(j+b)} + (2\pi)^5 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i+1}}{(2i+6)!} \\ \frac{2(\sin \pi b)^2}{(j+b)^6} = \frac{(2\pi)^2}{2!(j+b)^4} - \frac{(2\pi)^4}{4!(j+b)^2} + (2\pi)^6 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i}}{(2i+6)!} \\ \vdots \end{cases}$$

Likewise, for the sine, starting from,

$$\sin 2\pi(j+b) = \sin 2\pi b = \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i+1}}{(2i+1)!},$$

the recurrence equations below are obtained:

$$\begin{cases} \frac{\sin 2\pi b}{j+b} = 2\pi \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i}}{(2i+1)!} \\ \frac{\sin 2\pi b}{(j+b)^2} = \frac{2\pi}{j+b} - (2\pi)^2 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i+1}}{(2i+3)!} \\ \frac{\sin 2\pi b}{(j+b)^3} = \frac{2\pi}{(j+b)^2} - (2\pi)^3 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i}}{(2i+3)!} \\ \frac{\sin 2\pi b}{(j+b)^4} = \frac{2\pi}{(j+b)^3} - \frac{(2\pi)^3}{3!(j+b)} + (2\pi)^4 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i+1}}{(2i+5)!} \\ \frac{\sin 2\pi b}{(j+b)^5} = \frac{2\pi}{(j+b)^4} - \frac{(2\pi)^3}{3!(j+b)^2} + (2\pi)^5 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i}}{(2i+5)!} \\ \frac{\sin 2\pi b}{(j+b)^6} = \frac{2\pi}{(j+b)^5} - \frac{(2\pi)^3}{3!(j+b)^3} + \frac{(2\pi)^5}{5!(j+b)} - (2\pi)^6 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i+1}}{(2i+7)!} \\ \vdots \end{cases}$$

The formulae derived with these approaches break down for integer and half-integer b .

4.1 Cosine approach

Using the two systems of recurrence equations, we have more than one way to derive formulae for $\text{HP}_k(n)$. Unless they are combined, one has a very complicated recurrence to determine the formula for $\text{HP}_k(n)$. Let us go over the simplest way.

Setting $a = 1$ without loss of generality, and intertwining the two systems of equations, the cosine-based $\text{HP}_k(n)$ formula is,

$$\sum_{j=1}^n \frac{1}{(j+b)^k} = -\frac{1}{2b^k} + \frac{1}{2(n+b)^k} - \frac{(2\pi)^k}{2} \int_0^1 p_k(u) (\cos 2\pi(n+b)u - \cos 2\pi bu) \cot \pi u \, du,$$

and the simplified recurrence equation on $p_k(u)$ is below:

$$p_k(u) = \frac{(-1)^{\lfloor (k+1)/2 \rfloor}}{2 \sin \pi b} \left(\frac{(1-u)^{k-1}}{(k-1)!} + \sum_{j=1}^{\lfloor k/2 \rfloor} \frac{(-1)^j p_{2j-1}(u)}{(k-2j+1)!} \right) \cdot \begin{cases} -\frac{1}{\sin \pi b}, & \text{if } k \text{ is odd,} \\ \frac{1}{\cos \pi b}, & \text{if } k \text{ is even} \end{cases}$$

To solve this equation, it is necessary to separate odd from even. For the odd, one can make $q_k(u) = p_{2k+1}(u)$, and hence for $k \geq 0$:

$$q_k(u) = \frac{(-1)^k}{2(\sin \pi b)^2} \left(\frac{(1-u)^{2k}}{(2k)!} - \sum_{j=0}^{k-1} \frac{(-1)^j q_j(u)}{(2k-2j)!} \right)$$

The recurrence for even is very similar and is a function of $q_k(u)$. The generating functions of odd and even $p_k(u)$ are therefore,

$$f(x) = \frac{x \cos x(1-u)}{\cos x - \cos 2\pi b} \text{ and } g(x) = \sin x \frac{x \cos x(1-u)}{\cos x - \cos 2\pi b},$$

and the final formulae can be expressed as,

$$\begin{aligned} \sum_{j=1}^n \frac{1}{(j+b)^{2k+1}} &= -\frac{1}{2b^{2k+1}} + \frac{1}{2(n+b)^{2k+1}} \\ &\quad - \frac{(2\pi)^{2k+1}}{2} \int_0^1 \frac{f^{(2k+1)}(0)}{(2k+1)!} (\cos 2\pi(n+b)u - \cos 2\pi bu) \cot \pi u \, du, \end{aligned} \quad (23)$$

and

$$\begin{aligned} \sum_{j=1}^n \frac{1}{(j+b)^{2k}} &= -\frac{1}{2b^{2k}} + \frac{1}{2(n+b)^{2k}} \\ &\quad - \frac{(2\pi)^{2k}}{2 \sin 2\pi b} \int_0^1 \left(\frac{(-1)^k (1-u)^{2k-1}}{(2k-1)!} + \frac{g^{(2k)}(0)}{(2k)!} \right) (\cos 2\pi(n+b)u - \cos 2\pi bu) \cot \pi u \, du \end{aligned}$$

4.2 Sine approach

Since the process for sine is analogous, only the final formulae are shown. Let $f(x)$ and $g(x)$ be the functions,

$$f(x) = \frac{x \sin x(1-u)}{\cos x - \cos 2\pi b} \text{ and } g(x) = \sin x \frac{x \sin x(1-u)}{\cos x - \cos 2\pi b}$$

The even and odd $\text{HP}_k(n)$ formulae can be expressed as a function of their derivatives by,

$$\begin{aligned} \sum_{j=1}^n \frac{1}{(j+b)^{2k}} &= -\frac{1}{2b^{2k}} + \frac{1}{2(n+b)^{2k}} \\ &+ \frac{(2\pi)^{2k}}{2} \int_0^1 \frac{f^{(2k)}(0)}{(2k)!} (\sin 2\pi(n+b)u - \sin 2\pi bu) \cot \pi u \, du, \quad (24) \end{aligned}$$

and,

$$\begin{aligned} \sum_{j=1}^n \frac{1}{(j+b)^{2k+1}} &= -\frac{1}{2b^{2k+1}} + \frac{1}{2(n+b)^{2k+1}} \\ &+ \frac{(2\pi)^{2k+1}}{2 \sin 2\pi b} \int_0^1 \left(\frac{(-1)^k (1-u)^{2k}}{(2k)!} + \frac{g^{(2k+1)}(0)}{(2k+1)!} \right) (\sin 2\pi(n+b)u - \sin 2\pi bu) \cot \pi u \, du \end{aligned}$$

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